Exponentials and Logarithms

Questions 1–159

1.22 — Question 1. Solve the following equations, writing your answer in logarithmic form:

a) $4^x = 3$
b) $3^x = 4$
c) $2^x = \frac{1}{3}$
d) $9^x = \frac{1}{100}$

1.22 — Question 2. Solve the following equations, writing your answer in logarithmic form:

a) $2^{x+1} = 3$
b) $3^{x-5} = \frac{1}{5}$
c) $5^{2x-1} = 7^{500}$
d) $6^{3x+4} = 9^{250}$

1.22 — Question 3. Solve the equation

$$6^{x^2-5} = \frac{1}{216}$$

giving your answers in exact form.

1.22 — Question 4. Solve the equation

$$8^{10-x^2} = 32768$$

giving your answers in exact form.

1.22 — Question 5. Solve the following equations, writing your answer in the form $x = a^b$:

a) $\log_3 x = 8$
b) $\log_8 x = 3$
c) $\log_5 x = 7$
d) $\log_7 x = 5$

1.22 — Question 6. Write each of these as a single logarithm:

a) $\log_3 4 + \log_3 5$
b) $\log_5 3 + \log_5 6$
c) $\log_6 10 - \log_6 2$
d) $\log_{10} 4 - \log_{10} 8$

1.22 — Question 7. Write each of these as a single logarithm:

a) $\log_{10} 4 + \log_{10} 3 - \log_{10} 2$
b) $\log_2 20 - \log_2 6 + \log_2 3$
c) $\log_5 9 - \log_5 10 - \log_5 3$
d) $\log_9 8 - \log_9 10 + \log_9 4$

1.22 — Question 8. Solve the equation

$$\log_3 x - \log_3 5 = \log_3 8$$

1.22 — Question 9. Solve the equation

$$\log_7 9 - \log_7 x = \log_7 18$$

1.22 — Question 10. Evaluate each of these:

a) $\log_3 3$
b) $\log_8 8$
c) $2\log_5 5$
d) $-4\log_{10} 10$

1.22 — Question 11. Write each of these in the form $\log_a b$:

a) $2\log_3 5$
b) $3\log_4 3$
c) $4\log_9 2$
d) $5\log_7 10$

1.22 — Question 12. Write each of these in the form $\log_a b$, where b is a rational number:

a) $-\log_2 3$
b) $-\log_3 2$
c) $-2\log_5 2$
d) $-3\log_8 3$

1.22 — Question 13. Write each of these as a single logarithm:

a) $\log_b 2 + 2\log_b 5$

b) $\log_b 5 + 3\log_b 3$

c) $\log_b 10 - 4\log_b 2$

d) $\log_b 4 - 2\log_b 10$

1.22 — Question 14.

a) Express $3\log_2 x + \log_2 k$ as a single logarithm.

b) Given that $3\log_2 x + \log_2 k = 3$, express x in terms of k

1.22 — Question 15.

a) Express $5\log_3 x - \log_3 m$ as a single logarithm.

b) Given that $5\log_3 x - \log_3 m = 2$, express x in terms of m

1.22 — Question 16. Write each of these as a single logarithm:

a) $\log_5 30 - 2\log_5 2$

b) $\log_2 9 - 3\log_2 3$

c) $\log_9 100 + 2\log_9 5$

d) $\log_3 20 + 3\log_3 10$

1.22 — Question 17. Given that $\log_5 y^5 - \log_5 x^3 = 10$, show that $y = mx^n$, where m is an integer and n is a rational number.

1.22 — Question 18. Write each of these as a single logarithm:

a) $2\log_3 4 + 4\log_3 2$
b) $3\log_5 4 - 2\log_5 2$
c) $3\log_8 5 + 2\log_8 10$
d) $4\log_4 3 - 3\log_4 9$

1.22 — Question 19. Write each of these as a single logarithm:

a) $\frac{1}{2}\log_7 4 + \frac{1}{3}\log_7 8$
b) $\frac{1}{2}\log_{10} 81 - \frac{1}{3}\log_{10} 125$
c) $\frac{2}{3}\log_2 27 + \frac{3}{2}\log_2 9$
d) $\frac{4}{5}\log_9 32 - \frac{5}{4}\log_9 16$

1.22 — Question 20. Write each of these as a single logarithm:

a) $4\log_3 2 + 1$
b) $3\log_2 5 - 1$
c) $2\log_5 6 + 2$
d) $3 - 2\log_4 10$

1.22 — Question 21. Write each of these as a single logarithm:

a) $2\log_b 3 + 2$

b) $4\log_b 2 - 1$

c) $3\log_b 4 + 1$

d) $1 - 2\log_b 20$

1.22 — Question 22. It is given that

$$\log_b p - 2\log_b q = \log_b(p - 2q)$$

Show that

$$p = \frac{2q^3}{q^2 - 1}$$

where $p > 2q > 0$

1.22 — Question 23. It is given that

$$3\log_b q - \frac{1}{2}\log_b p = \log_b(3q - p)$$

Show that

$$p = \frac{q^6}{9q^2 - 6pq + p^2}$$

where $3q > p > 0$

1.22 — Question 24. Find the value of:

a) $\log_b b^2 + \log_b b^5$
b) $\log_b b^4 - \log_b \frac{1}{b}$
c) $\log_b \frac{1}{b^2} + \log_b b^3$
d) $\log_b \frac{1}{b} - \log_b \frac{1}{b^3}$

1.22 — Question 25. Show that, for $x > 0$,

$$\log_3 \frac{x^3}{9} + \log_3 27x - \log_3 x^5 = \log_3 \frac{3}{x}$$

1.22 — Question 26. Given that $m = \log_4 x$, where x is positive, write each of these in their simplest form in terms of m:

a) $\log_4 \sqrt{x}$
b) $\log_4 \frac{x}{4}$
c) $\log_4 \frac{16}{x^2}$

1.22 — Question 27. It is given that

$$\log_b y = 3\log_b 2 + \log_b 5 - \frac{3}{2}$$

Write y in terms of b without any logarithms.

1.22 — Question 28. It is given that

$$\log_k y = \log_k 8 - 2\log_k \frac{1}{2} + \frac{3}{4}$$

Write y in terms of k without any logarithms.

1.22 — Question 29. Write each of these as a single logarithm:

a) $2\log_b p + 3\log_b q + 2\log_b r$
b) $3\log_b p + 5\log_b q - 3\log_b r$
c) $5\log_b p - 2\log_b q + 4\log_b r$
d) $8\log_b p - 6\log_b q - 9\log_b r$

1.22 — Question 30. Write each of these as a single logarithm:

a) $\log_7 30 - (\log_7 2 + \log_7 3)$
b) $\log_8 15 - 2(\log_8 6 - \log_8 2)$
c) $\log_6 60 - 3(\log_6 5 + \log_6 2)$
d) $\log_{10} 50 - 2(\log_{10} 80 - \log_{10} 16)$

1.22 — Question 31. Write each of these as a single logarithm:

a) $\log_b p - (\log_b q + \log_b r)$
b) $\log_b p - 2(\log_b q - \log_b r)$
c) $\log_b p - 3(\log_b q + \log_b r)$
d) $2\log_b p - 2(\log_b q - 2\log_b r)$

1.22 — Question 32. It is given that $x = \log_b 3$, $y = \log_b 5$ and $z = \log_b 7$. Write each of these in terms of x, y and z:

a) $\log_b 15$
b) $\log_b 9$
c) $\log_b 21$
d) $\log_b \frac{49}{25}$

1.22 — Question 33. By writing $\log_3 4 \times \log_3 64$ in terms of $\log_3 4$, show that

$$\log_3 4 \times \log_3 64 > 3$$

1.22 — Question 34. Showing your working, write each of these in a form not involving logarithms:

a) $\log_{100} 10 + \log_{10} 100$
b) $\log_2 8 - \log_8 2$
c) $\log_9 81 + \log_3 \frac{1}{3}$
d) $\log_{25} \frac{1}{625} + \log_5 625$

1.22 — Question 35. A right-angled triangle ABC has hypotenuse AC. The length of AB is $(\log_5 16)$ cm and the length of BC is $(\log_5 64)$ cm. Without the use of a calculator, show that the area of the triangle is larger than 0.75 cm^2.

1.22 — Question 36. Using $y = 2^x$ as a substitution, show that

$$4^x - 7 \times 2^{x+1} + 45 = 0$$

can be written as

$$y^2 - 14y + 45 = 0$$

Hence show that $4^x - 7 \times 2^{x+1} + 45 = 0$ has two solutions of the form $\log_2 k$.

1.22 — Question 37. By writing $\log_4 3 \times \log_4 9 \times \log_4 81$ in terms of $\log_4 3$, show that

$$\log_4 3 \times \log_4 9 \times \log_4 81 < 8$$

1.22 — Question 38. Using $y = 2^{2x}$ as a substitution, show that

$$16^x + 2^{2x+1} - 99 = 0$$

can be written as

$$y^2 + 2y - 99 = 0$$

Hence show that $16^x + 2^{2x+1} - 99 = 0$ has only one solution of the form $\log_2 k$.

1.22 — Question 39. Solve the following equations, writing your answers in logarithmic form:

a) $2^{2x} - 8 \times 2^x + 15 = 0$

b) $3^{2x} - 9 \times 3^x + 20 = 0$

c) $4^{2x} - 6 \times 4^x - 7 = 0$

d) $3^{2x} - 6 \times 3^x - 40 = 0$

1.22 — Question 40. Solve the following equations, writing your answers in logarithmic form:

a) $2^{2x} - 5 \times 2^{x+1} + 21 = 0$

b) $5^{2x} - 3 \times 5^{x+1} + 50 = 0$

c) $4^{2x} + 2^{2x+1} - 15 = 0$

d) $4^{2x} - 5 \times 2^{2x+1} + 2^{2x+3} - 80 = 0$

1.22 — Question 41. Solve the equation

$$8^x - 2^{2x+2} + 3 \times 2^x = 0$$

1.22 — Question 42. Solve the following equations, writing your answers in logarithmic form:

a) $3^{2x+1} - 13 \times 3^x - 10 = 0$

b) $16^{x+1} - 139 \times 4^x - 45 = 0$

c) $4^{2x+3} - 9 \times 2^{2x+5} = 0$

d) $2^{4x+7} + 2^{2x+3} - 3 \times 4^{x+2} - 3 = 0$

1.22 — Question 43. Show that the solution to the equation

$$5^{x-3} = 2^{x+4}$$

can be written in the form

$$\frac{4\log_5 2 + 3}{1 - \log_5 2}$$

1.22 — Question 44. Show that the solution to the equation

$$8^{x+2} = 3^{2-x}$$

can be written in the form

$$\frac{2 - 2\log_3 8}{\log_3 8 + 1}$$

1.22 — Question 45. Show that the solution to the equation

$$3^{x+1} = 8^{x-1}$$

can be written in the form

$$\frac{\log_{10} 8 + \log_{10} 3}{\log_{10} 8 - \log_{10} 3}$$

1.22 — Question 46. Solve each of these equations by taking logarithms of base 3 of both sides:

a) $3^x = 4^{x-1}$
b) $3^{x+5} = 5^{2-x}$
c) $8^{x-3} = 3^{x+8}$
d) $10^{5-x} = 3^{x-6}$

1.22 — Question 47. Solve each of these equations by taking logarithms of base 2 of both sides:

a) $5^x = 9^{x+2}$

b) $3^{x-2} = 7^{3-x}$

c) $9^{x+4} = 10^{4-x}$

d) $5^{8-x} = 6^{4-x}$

1.22 — Question 48.

a) Given that

$$2\log_{10}(2-x) = \log_{10}(28-9x)$$

show that $x^2 + 5x - 24 = 0$

b) Hence show that

$$2\log_{10}(2-x) = \log_{10}(28-9x)$$

has only one solution.

1.22 — Question 49. Solve the equation

$$\log_5(8x-1) - \log_5(2-x) = 2$$

1.22 — Question 50. Julia is trying to solve the equation

$$3(\log_8 x)^2 + 11\log_8 x - 20 = 0$$

Julia's attempt is shown below:

$$\begin{aligned} 3(\log_8 x)^2 + 11\log_8 x - 20 &= 0 && \text{Line 1} \\ 6\log_8 x + 11\log_8 x - 20 &= 0 && \text{Line 2} \\ 17\log_8 x - 20 &= 0 && \text{Line 3} \\ \log_8 x &= \tfrac{20}{17} && \text{Line 4} \\ x &= (\tfrac{20}{17})^8 && \text{Line 5} \end{aligned}$$

a) Identify the two errors made by Julia.
b) Solve the equation

$$3(\log_8 x)^2 + 11\log_8 x - 20 = 0$$

giving your answers in exact form.

1.22 — Question 51. George is trying to solve the equation

$$\log_3 \frac{1}{x} + \frac{1}{4\log_3 x} = 0$$

George's attempt is shown below:

$$\log_3 \tfrac{1}{x} + \tfrac{1}{4\log_3 x} = 0 \quad \text{Line 1}$$
$$\log_3 \tfrac{1}{x} + \tfrac{1}{4}\log_3 \tfrac{1}{x} = 0 \quad \text{Line 2}$$
$$\tfrac{5}{4}\log_3 \tfrac{1}{x} = 0 \quad \text{Line 3}$$
$$-\tfrac{5}{4}\log_3 x = 0 \quad \text{Line 4}$$
$$x = 3 \quad \text{Line 5}$$

a) Identify the two errors made by George.

b) Solve the equation

$$\log_3 \frac{1}{x} + \frac{1}{4\log_3 x} = 0$$

giving your answers in exact form.

1.22 — Question 52.

a) Show that the equation

$$2\log_3 x = \log_3(mx - 4) + 4$$

where m is a positive constant, can be expressed in the form

$$x^2 - 81mx + 324 = 0$$

b) Given that $x^2 - 81mx + 324 = 0$ has only one root, find the value of m

1.22 — Question 53. Solve $2^{2^x} = 5$, writing your answer in an exact form.

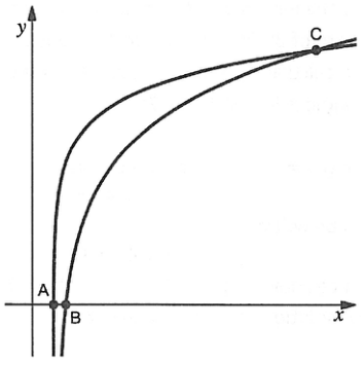
1.22 — Question 54. The graph below shows two curves that intersect at the point C. The curve that passes through A has equation

$$y = \log_5(6x - 15) + 1$$

and the curve that passes through B has equation

$$y = 2\log_5(x - 3)$$

Determine the exact coordinates of the points A, B and C.



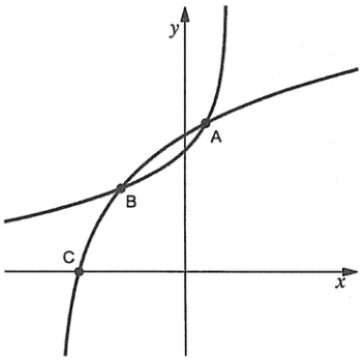
1.22 — Question 55. The graph below shows two curves that intersect at the points A and B. The curve that passes through C has equation

$$y = 2\log_4(x+6)$$

and the other curve has equation

$$y = \log_4(x+48) - \log_4(2-x)$$

Determine the exact coordinates of the points A, B and C.



1.23 — Question 1. Solve the following equations, writing your answer in logarithmic form:

a) $e^x = 8$
b) $e^{2x} = 4$
c) $e^{3x} = 8$
d) $e^{5x} = \frac{1}{100}$

1.23 — Question 2. Solve each of these equations:

a) $e^{4x+3} = 9$

b) $e^{5x-2} = 10$

c) $e^{9-2x} = 4$

d) $e^{1-5x} = \frac{1}{5}$

1.23 — Question 3. Solve the following equations, writing your answer in the form $x = e^a$.

a) $\ln x = 7$

b) $\ln x = -5$

c) $\ln x = \frac{1}{3}$

d) $\ln x = -\frac{2}{5}$

1.23 — Question 4. Write each of these as a single logarithm:

a) $\ln 4 + \ln 5 + \ln 6$
b) $\ln 8 + \ln 3 - \ln 4$
c) $\ln 12 - \ln 4 - \ln 6$
d) $\ln 40 - \ln 8 + \ln 5$

1.23 — Question 5. Solve the following equations:

a) $\ln x + \ln 2 = \ln 8$
b) $\ln x - \ln 5 = \ln 3$
c) $\ln 8 + \ln x = \ln \frac{1}{3}$
d) $\ln 50 - \ln x = \ln \frac{3}{2}$

1.23 — Question 6. Evaluate each of these:

a) $\ln e$
b) $4 \ln \frac{1}{e}$
c) $\ln \frac{1}{e^5}$
d) $\ln \sqrt[3]{e}$

1.23 — Question 7. Write each of these as a single logarithm:

a) $2 \ln 2 + 2 \ln 4$
b) $3 \ln 2 - 2 \ln 3$
c) $2 \ln 9 + 3 \ln 3 - 2 \ln 6$
d) $2 \ln 10 - 4 \ln 2 - 2 \ln 5$

1.23 — Question 8. Write $3\ln x - 4\ln y$ as a single logarithm.

1.23 — Question 9. Write each of these as a single logarithm:

a) $4\ln p + 2\ln q + 3\ln r$
b) $3\ln p + 5\ln q - 4\ln r$
c) $6\ln p - 2\ln q - 8\ln r$
d) $9\ln p - 5\ln q + 7\ln r$

1.23 — Question 10. Write each of these as a single logarithm:

a) $2\ln p - (\ln q + 3\ln r)$

b) $\ln p - (2\ln q - \ln r)$

c) $3\ln p - 2(\ln q + 2\ln r)$

d) $4\ln p - 3(2\ln q - 3\ln r)$

1.23 — Question 11. Write each of these equations in a form not involving logarithms:

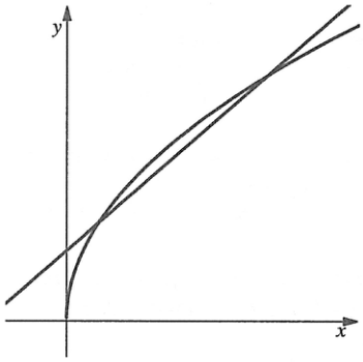
a) $\ln x + \ln y = 5$

b) $\ln y - \ln x = 12$

c) $2\ln x - \ln y = 4$

d) $4\ln x + 3\ln y = 8$

1.23 — Question 12. The diagram below shows the straight-line graph $7y = x + 10$ and the curve with equation $\ln x = 2 \ln y$. Determine the exact coordinates of where the two graphs intersect.



1.23 — Question 13. Solve each of these equations:

a) $e^{2x} - 16e^{x} = 0$
b) $e^{3x} - 16e^{x} = 0$
c) $e^{3x} - 16e^{2x} = 0$
d) $e^{4x} - 16e^{2x} = 0$

1.23 — Question 14. Solve each of these equations:

a) $e^x = e^{2x+1}$

b) $e^{2x} = e^{3x-5}$

c) $e^x = 2e^{4x}$

d) $e^{2x} = 3e^{x-1}$

1.23 — Question 15.

a) Using $y = e^x$ as a substitution, show that

$$4e^x + 27e^{-x} - 39 = 0$$

can be written as

$$4y^2 - 39y + 27 = 0$$

b) Hence solve the equation

$$4e^x + 27e^{-x} - 39 = 0$$

giving your answers as exact values.

1.23 — Question 16. Solve the following equations, writing your answers in logarithmic form:

a) $e^{2x} - 6e^x + 8 = 0$

b) $e^{2x} - 13e^x + 40 = 0$

c) $e^x + 4 - 45e^{-x} = 0$

d) $e^x + 5 - 150e^{-x} = 0$

1.23 — Question 17. Solve the following equations, writing your answers in logarithmic form:

a) $2e^{2x} - 7e^x + 6 = 0$

b) $4e^{2x} - 28e^x + 45 = 0$

c) $6e^x - 23 - 18e^{-x} = 0$

d) $8e^x + 14 - 15e^{-x} = 0$

1.23 — Question 18. Show that the solution to the equation

$$4^{x-3} = 5^{x+6}$$

can be written in the form

$$\frac{6\ln 5 + 3\ln 4}{\ln 4 - \ln 5}$$

1.23 — Question 19. By writing $\ln 5 \times \ln 125 \times \ln 625$ in terms of $\ln 5$, show that

$$\ln 5 \times \ln 125 \times \ln 625 > 12$$

1.23 — Question 20. Solve each of these equations by taking natural logarithms of both sides:

a) $2^x = 5^{x+4}$

b) $4^{x-1} = 6^{x+2}$

c) $5^{3-x} = 7^{x+1}$

d) $8^{9-x} = 6^{8-x}$

1.23 — Question 21. Solve each of these equations, writing your answers in exact form:

a) $2\ln(3x-5) - 4 = 0$

b) $2\ln(3-2x) - 9 = 0$

c) $2\ln(x+4) - \ln 5 = 0$

d) $5\ln 2x - 3\ln 3x = 1$

1.23 — Question 22. Solve the equation $2\ln x - \ln(x+12) = 0$

1.23 — Question 23. Solve each of these equations, writing your answers in exact form:

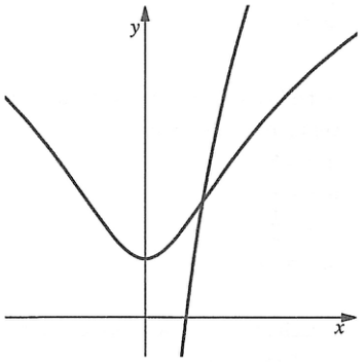
a) $\ln x = 2 + \ln(x+3)$

b) $\ln(x-1) = 3 + \ln(x+1)$

c) $\ln 2x = 4 - \ln x$

d) $\ln 4x = 5 - \ln(x+1)$

1.23 — Question 24. The diagram below shows the curves $y = \ln(x^2+2)$ and $y = 4\ln x$. Find the exact coordinates of where the two curves intersect.



1.23 — Question 25. Solve each of these equations, writing your answers in exact form:

a) $2\ln x = 2 + \ln 2x$
b) $2\ln x = 3 - \ln 8x$
c) $2\ln x = \ln 4x - 1$
d) $2\ln x = \ln 10x - 4$

1.23 — Question 26. Solve the equation

$$3\ln x = 3 - \ln 16x$$

writing your answer in exact form.

1.23 — Question 27. Solve each of these equations, writing your answers in exact form:

a) $\ln x + \frac{32}{\ln x} = 12$
b) $2\ln x + \frac{6}{\ln x} = 7$
c) $6\ln x + 11 = \frac{10}{\ln x}$
d) $10\ln x - 33 = \frac{7}{\ln x}$

1.23 — Question 28. Solve the equation

$$2\ln(x+10) = \ln x + \ln(2x-1)$$

1.23 — Question 29. Solve the equation

$$\ln(5-3x) + \ln(8+4x) = 2\ln(x+2)$$

1.23 — Question 30. Solve the equation

$$2\ln(5-2x) = \ln 2 + \ln(\frac{233}{8} - 2(x-\frac{5}{4})^2)$$

1.23 — Question 31.

a) Using $y = e^x$ as a substitution, show that

$$1 - 28e^{-3x} - 4e^{-2x} + 7e^{-x} = 0$$

can be written as

$$y^3 + 7y^2 - 4y - 28 = 0$$

b) Hence solve the equation

$$1 - 28e^{-3x} - 4e^{-2x} + 7e^{-x} = 0$$

giving your answers as exact values.

1.23 — Question 32. Given that $y = \ln x$ and $y^2 - 3y + 2 = 0$, find the possible values of x

1.23 — Question 33. Solve the equation

$$6(\ln x)^2 - 19\ln x + 15 = 0$$

1.23 — Question 34. Solve the equation

$$\ln 2x + \ln 2x^2 + \ln 2x^3 = 6$$

1.23 — Question 35. Solve the equation

$$\frac{e^x}{3^x} = e$$

writing your answer in the form

$$\frac{p}{q + \ln r}$$

where p, q and r are integers.

1.23 — Question 36. Solve the equation

$$3^x e^{2x+3} = 2$$

writing your answer in the form

$$\frac{p + \ln q}{r + \ln s}$$

where p, q, r and s are integers.

1.23 — Question 37. Solve the equation

$$5^x e^{8x} = e^{10}$$

writing your answer in the form

$$\frac{p}{q + \ln r}$$

where p, q and r are integers.

1.23 — Question 38. Fully simplify

$$\frac{3x^2 - 19x + 6}{x^2 - x - 30}$$

Hence, given that

$$\ln(3x^2 - 19x + 6) - 1 = \ln(x^2 - x - 30)$$

where $x \neq 6$, find x in terms of e.

1.23 — Question 39. Determine the exact coordinates of where the curves $y = 2e^x - 11$ and $y = 40e^{-x}$ intersect.

1.23 — Question 40. A curve has equation $y = ae^x + b$ and passes through the points $(\ln 2, 12)$ and $(-3\ln 2, \frac{87}{16})$. Find the values of a and b

1.23 — Question 41. A curve has equation $y = p\ln x + q$ and passes through the points $(e^{12}, -\frac{55}{2})$ and $(e, \frac{11}{6})$. Find the values of p and q

1.23 — Question 42. Solve the simultaneous equations

$$e^x + y^2 = 4$$

$$2\ln y = x$$

1.23 — Question 43. Determine the exact coordinates of where the curves $x + e^y = 4$ and $\ln((x-4)^2) - y = 2$ intersect.

1.23 — Question 44. The water level, h m, in a large tank over time, t hours, is modelled by the equation

$$h = 20 - 5\ln(t+1)$$

a) What is the initial height of the water in the tank?
b) How high is the water in the tank after 1 hour?
c) After how many hours will the water level be 10 metres?

1.23 — Question 45. A population of hawks, P, can be modelled by the equation

$$P = 100\ln(\frac{t+3}{3}) + 500$$

where t is the number of years since a conservation project started.
a) What was the initial population of hawks when the conservation project started?
b) What does the model suggest the size of the population will be in 5 years after the conservation project started?
c) After how many years does the model suggest the population will reach 700 hawks?

1.23 — Question 46. The temperature, $T°$C, of the water in a lake can be modelled by the equation

$$T = 2\ln(t+3) - \ln(t+2) + \ln 100$$

where t is the number of days after the 1st March.

a) What was the temperature of the lake on the 1st March?

b) How many days after the 1st March will the temperature of the lake be $(\ln 3000)°$C?

1.26 — Question 1. For each of the equations below find the value of y when $x = 0$:

a) $y = 1 + 2^{-x}$

b) $y = 20 - 2^{0.1x}$

c) $y = 10 + e^{-0.1x}$

d) $y = 5 - e^{0.001x}$

e) $y = 21 + 5e^{-0.5x}$

f) $y = 3 - 5 \times (\frac{1}{2})^{-\frac{1}{4}x}$

1.26 — Question 2. Find the value of P when $t = 0$ for the following:

a) $P = \frac{100}{1+e^{-\frac{1}{2}t}}$

b) $P = \frac{18}{1+e^{-0.1t}}$

1.26 — Question 3. A population of bacteria is modelled with the equation

$$P = 250e^{0.15t}$$

where P is the population size and t is the time in hours.

a) State the size of the initial population.

b) Find the size of the population after 20 hours.

c) When does the population reach 1000 bacteria?

1.26 — Question 4. For each of the models below, find the time t (hours) it takes for an amount of a substance y (grams) to double from its initial amount, giving your answers in exact form.

a) $y = 3 \times 2^t$
b) $y = 4e^{1.5t}$
c) $y = 150 \times 1.6^t$
d) $y = 20 + 4e^{0.1t}$

1.26 — Question 5. The quantity of radioactive isotope Uranium-235 y grams over time may be modelled by the equation

$$y = 1000 \times 2^{-0.00142t}$$

where t is millions of years.

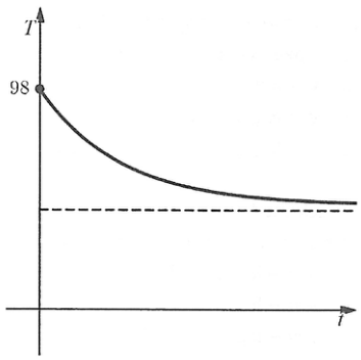
a) State the initial mass of the Uranium-235
b) Show that the half-life of Uranium-235 is about 704 million years.

1.26 — Question 6. For each of the models below, find the initial value of y and state what is happening to the value of y as $x \to \infty$
Hence sketch each of the models.

a) $y = 10 + 2e^{-0.3x}$
b) $y = -5 + 7e^{-2x}$
c) $y = 1.2 - 0.6e^{-7.6x}$
d) $y = 0.2 + 1.8e^{2.3x}$

1.26 — Question 7. The graph below shows the temperature of a cup of coffee over time.
The room temperature is 22°C.
The temperature, T°C, after t minutes may be modelled by the equation

$$T = A + Be^{-\frac{1}{5}t}$$



a) State the initial temperature of the coffee.
b) Write down the equation of the horizontal asymptote.
c) Hence find the values of A and B
d) Suggest how the model may be refined for a cup of coffee that is initially 100°C.
e) Suggest values of A and B for a cup of coffee cooling at the same rate but from initial temperature 97°C in a room at 18°C.

1.26 — Question 8. The height of a plant is given by the equation

$$h = \frac{50}{1 + 4e^{-\frac{1}{5}t}}$$

where h is the height in cm and t is the number of weeks that have passed.

a) According to this model state the initial height of the plant.

b) What height does this model predict in the long term for this plant?

1.26 — Question 9. The height of a child was recorded for the first seven years of their life. A model suggested for the child's height, h cm, at x years old is given by

$$h = 130 - 80 \times 1.45^{-x}$$

where $0 \leq x \leq 7$. Using the model:

a) State the height of a newborn baby.

b) Find the height of a child after 18 months, giving your answer to the nearest millimetre.

c) Estimate the age of the child in months (to the nearest month) when their height is 1 m.

d) Explain why this model might be unsuitable for predicting the child's height as a teenager.

1.26 — Question 10. An investment of £1200 is made at 5% compound interest per annum, with interest added to the balance only at the end of each complete year.

a) Create an equation to model the value of the investment over time.
b) Sketch the curve for your model, including labelling any points of intersections with axes.
c) Find the value of the investment after 4 years giving your answer to the nearest penny.
d) Find when the investment reaches more than £2000.

1.26 — Question 11. A fine for speeding is initially £A. Each week it is not paid, the fees increase exponentially. After 1 week the fine becomes £150 and after 2 weeks it becomes £225.

a) Calculate the initial fine, £A
b) Find a model which for this data, giving your answer in the form

$$C = A \times r^t$$

where C is the cost, t is the time (in weeks) and A and r are constant values to be found.
c) Use your model to find the cost of the fine in week 4.

1.26 — Question 12. The number of people using a social media platform is declining exponentially since its peak of popularity (at $t = 0$).
At 3 years and 6 years it had 180 and 144 million active users respectively.

a) Find the number of active users at its peak.

b) Hence find a model which fits the data, giving your answer in the form

$$N = A \times r^{\frac{1}{3}t}$$

where N is the number of active users, in millions, t is the time, in years, and A and r are constant values to be found.

c) Use your model to predict the number of users after 10 years has passed.

1.26 — Question 13. A population of insects is known to be growing exponentially. The population 2 and 4 months into the data collection is 1440 and 3240 insects respectively.
A model for the data may be given by

$$P = A \times r^t$$

where P is population, t is time (in months) and A and r are constant values to be found.

a) Find the values of A and r in the model.

b) Interpret the value of the constants A and r in context.

1.26 — Question 14. The amount of stock in a warehouse of a particular product P, is known to be 12 000 units. After 5 months this is expected to have reduced by 9500.

a) Find a model of the form

$$P = ae^{kt}$$

where t is the number of months passed, where a and k are constant values to be found.

b) How much stock is available after 2 months?

c) The company orders a delivery from the manufacturers when the stock drops to 500 units. After how many months should they make this order?

d) What does this model predict in the long term for the stock of this product? Comment on its reliability.

1.26 — Question 15. The amount, C milligrams, of a particular drug in a person's body after t hours may be given by a function of the form

$$C = Ae^{-kt}$$

a) Given that the initial dose taken is 200mg and in 4 hours the amount of drug has halved once, find the values of A and k giving your answer in exact form.

b) Hence, find the number of hours it takes for the drug to reduce to 50 mg in a person's body.

1.26 — Question 16. The number P of restaurants in a particular fast food chain in their nth year may be modelled by the function

$$P = Ae^{kn}$$

In the first year ($n = 1$), there were 1000 restaurants and in the second year there were 1400.

a) Show that $e^k = 1.4$
b) Find the exact value of k and A.
c) How many restaurants does the model predict in the 4th year?
d) How many whole years is it before there are 100 000 restaurants?
e) Explain why the model is invalid in the long-term.

1.26 — Question 17. A substance p is known to decay exponentially according to the model

$$p = p_0 e^{-kt}$$

where t is time in years, p_0 is the initial quantity and k is an unknown constant.
Find the amount of time needed for the quantity to have reduced by 99%, giving your answer in the form $\frac{1}{k} \ln A$ where A is a positive integer.

1.26 — Question 18. The amount of a drug y mg in the blood stream is modelled by the equation

$$y = Ae^{-kt}$$

where t is the time in hours since a tablet was taken. After 2 hours there was 500mg in the blood stream, and after 7 hours there was 88mg in the blood stream.

a) Find the exact value of k

b) Find the value of A to 1 decimal place.

1.26 — Question 19. The populations of two types of fish in a lake may be modelled with the following equations:

Fish A:

$$P_A = ae^{0.08t}$$

Fish B:

$$P_B = be^{0.12t}$$

where P_A and P_B are the respective populations of fish A and fish B and t is the number of months passed since the start of the year.

The total number of fish in the lake at the start of the year is 85, then after 6 months passes the total population is 153.

a) Find the values of a and b giving your answers to 1 decimal place.

b) Interpret what your values for a and b mean in context.

c) Find the number of months passed when the population of fish B first exceeds the population of fish A.

d) Explain why the models are invalid in the long-term.

1.26 — Question 20. A beaker of liquid is cooling. Its initial temperature is 65°C.
After 3 minutes its temperature is 35°C.
The temperature is modelled using the equation

$$T = 17 + ae^{-kt}$$

where T is the temperature in °C, t is the time in minutes and a and k are constants.

a) Find the exact values of a and k.
b) What is the temperature of the liquid after 5 minutes?
c) What is the temperature of the liquid in the long term?
d) State how the model could be modified if the room temperature is reduced.

The temperature of a second liquid, y°C, cooling from 80°C at the same point in time in the same room is modelled by the equation

$$y = A + be^{-1.5kt}$$

where k has the same value from before.

e) Find the time at which both liquids have the same temperature.

1.26 — Question 21. Prerequisite knowledge for Q21-Q26: Differentiation of the form ae^{kx} For each of the equations calculate the rate of change in y when $x = 2$:

a) $y = 20 + 2e^{-0.3x}$
b) $y = 4(5 + e^{0.2x})$
c) $y = 70 - 60e^{-0.8x}$

1.26 — Question 22. The population of red squirrels in the UK can be modelled by the equation

$$P = 3500e^{-\frac{1}{225}\ln(\frac{700}{57})t}$$

where P is their population (in thousands) and t is the number of years since 1800.
Show that in 2000, according to the model, the amount of squirrels is decreasing at a rate of approximately 4.2 thousand per year.

1.26 — Question 23. A volume V (ml) of liquid being poured from a bottle may be modelled by the equation

$$V = 330(1 - e^{-0.1t})$$

after t seconds.

a) State the theoretical maximum amount of liquid the bottle is holding, according to the model.
b) Find the rate at which the liquid is being poured when $t = 5$ seconds.

1.26 — Question 24. The amount of stock in a shop of two products P may be modelled by the following equations where t is the number of months since the start of the year.

Product A: $P = 100(1 + e^{-kt})$

Product B: $P = 400e^{-kt}$

where $t \geq 0$ and k is a positive constant.

a) Sketch the two models on the same axes, include labelling points where the curves meet the axes and any asymptotes.

b) State the amount of each product in the long term according to the model.

c) Explain the limitations of the model.

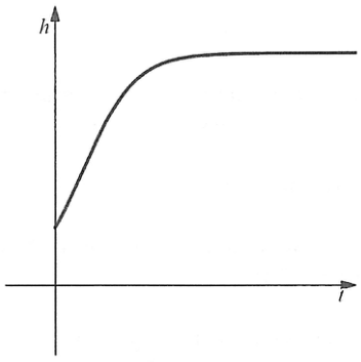
d) Show that the rate of decrease of product B is always 4 times that of product A.

e) Find the exact value of k given there is the same amount of each product after two months have passed.

1.26 — Question 25. The height of a tree is given by the equation

$$h = \frac{18e^{\frac{1}{2}t}}{e^{\frac{1}{2}t} + 3}$$

where h is the height in metres and t is the number of years that have passed since it is planted. The graph of height against time is shown below.



a) According to this model state the initial height of the tree.

b) Show that

$$h = \frac{18}{1 + 3e^{-\frac{1}{2}t}}$$

c) What height does this model predict in the long term for this tree?

1.26 — Question 25 (continued). A second tree was planted at the same time, and it may be modelled with equation

$$h = a - 14e^{-\frac{1}{2}t}$$

where a is an unknown constant.

d) Find the value of a given this tree has a height of 2 m when planted.

e) Find the rate of growth of the second tree at 5 years.

f) State which of the two trees is expected to be taller.

1.26 — Question 26. Two models of mass m kg over an exponentially decaying substance over time t days are given by

$$m = 100e^{-0.05t}$$

and

$$m = 10 + 90e^{-kt}$$

where k is a positive constant.

Find the ranges of values of k for which the second model is initially decaying at a faster rate than the first.

1.27 — Question 1. Answer sheets for questions marked with a (*) can be found at TLMaths.com
Show that $y = ax^b$ may be written in the form

$$\log_{10} y = b \log_{10} x + \log_{10} a$$

1.27 — Question 2. Show that $y = a \times b^x$ may be written in the form

$$\log_{10} y = x \log_{10} b + \log_{10} a$$

1.27 — Question 3. Show that $y = ae^{bx}$ may be written in the form

$$\ln y = bx + \ln a$$

1.27 — Question 4. A model for some data is given as

$$y = 2 \times 1.05^x$$

Show that the graph of $\log_{10} y$ against x will be a straight line.

1.27 — Question 5. A model for some data is given as

$$y = 0.5x^{1.2}$$

Show that the graph of $\ln y$ against $\ln x$ will be a straight line.

1.27 — Question 6. A model for some data is given as

$$y = 0.4 \times 0.88^{-x}$$

Show that the graph of $\log_{10} y$ against x will be a straight line.

1.27 — Question 7. For each of the following, rearrange to find an equation of y in terms of x, giving your answer in the form $y = ax^b$ where a and b are values given to 2 significant figures:

a) $\log_{10} y = 0.478 + 5 \log_{10} x$
b) $\ln y = 3 \ln x + 0.815$
c) $\log_{10} y = -1.347 - 0.89 \log_{10} x$
d) $\ln y = 0.51 \ln x - 1.15$

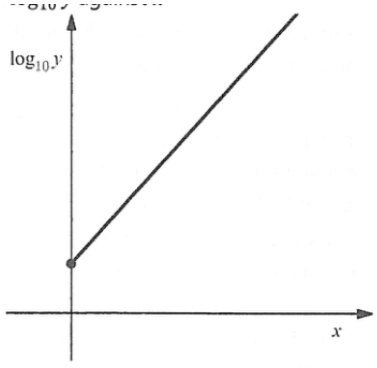
1.27 — Question 8. For each of the following, rearrange to find an equation of y in terms of x, giving your answer in the form $y = a \times b^x$ where a and b are values given to 2 significant figures:

a) $\log_{10} y = 0.214x - 1.54$
b) $\ln y = 0.555x + 0.444$
c) $\ln y = -0.235x + 0.159$
d) $\log_2 y = 0.895x + 2.03$

1.27 — Question 9. Convert each of the following into the form $\ln y = a + bx$ where a and b are constants to be found:

a) $y = 40e^{-0.2x}$
b) $y = 0.5 \times 1.04^{0.5x}$
c) $y = 1.62 \times 2.13^{-0.2x}$

1.27 — Question 10. A data set (x, y) has been plotted on a graph of $\log_{10} y$ against x



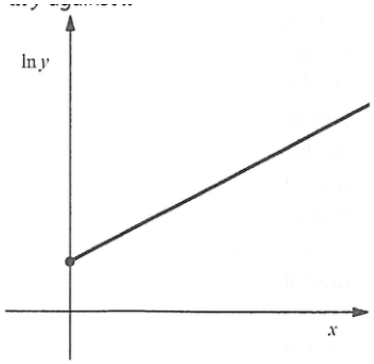
A linear equation is found to best fit the data. In each of the following cases, find a model in the form

$$y = a \times b^x$$

where a and b are values to be found:

a) gradient of 3 and an intercept of 0.5
b) gradient of 0.25 and an intercept of 0.0714

1.27 — Question 11. A data set (x, y) has been plotted on a graph of $\ln y$ against x



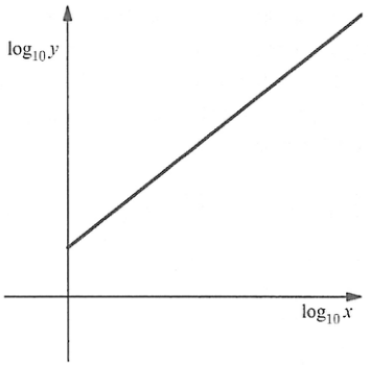
The linear equation found to best fit the data. In each of the following cases, find a model in the form

$$y = ae^{bx}$$

where a and b are values to be found:

a) gradient of 4 and an intercept of 2
b) gradient of 0.75 and an intercept of 0.1

1.27 — Question 12. A data set (x, y) has been plotted on a graph of $\log_{10} y$ against $\log_{10} x$



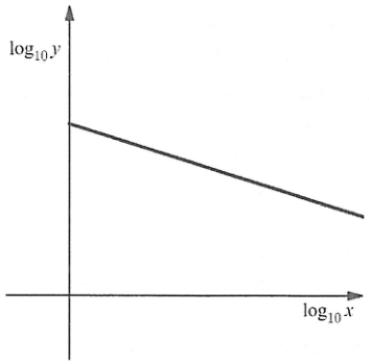
The linear equation found to best fit the data. In each of the following cases, find a model in the form

$$y = ax^n$$

where a and n are values to be found:

a) gradient of 4 and an intercept of 0.1
b) gradient of 0.5 and an intercept of 4

1.27 — Question 13. A data set (x, y) has been plotted on a graph of $\log_{10} y$ against $\log_{10} x$ The linear equation found to best fit the data.



In each of the following cases, find a model in the form

$$y = ax^n$$

where a and n are values to be found:

a) gradient of -5 and an intercept of 0.12
b) gradient of -0.4 and an intercept of 1.17

1.27 — Question 14. A straight-line graph of $\ln y$ against $\ln x$ passes through the y-axis at 2 and the point with coordinates $(1.8, 2.9)$. Find the equation which represents the equation of the line in terms of y and x

1.27 — Question 15. A straight-line graph of $\log_{10} y$ against $\log_{10} x$ passes through the y-axis at 3 and the point with coordinates $(1.6, 3.704)$. Find the equation which represents the equation of the line in terms of y and x

1.27 — Question 16. A straight-line graph of $\log_{10} y$ against x passes through the y-axis at 0.5 and the point with coordinates $(1.01, 0.803)$. Find the equation which represents the equation of the line in terms of y and x

1.27 — Question 17. A straight-line graph of $\ln y$ against x passes through the points $(1.1, 3.06)$ and $(2.4, 4.12)$. Find the equation which represents the equation of the line in terms of y and x

1.27 — Question 18. (*) The number of visits to a website y since its launch x days before is recorded in the following table:

Days since launch, x	Number of visits, y
2	5
30	15
60	24
100	30
150	40

It is thought that the data may be modelled in the form

$$y = ax^b$$

a) Complete a table of values of $\log_{10} x$ and $\log_{10} y$, giving your answers to 3 decimal places.
b) Plot a graph of $\log_{10} y$ against $\log_{10} x$ and draw a line of best fit on your plot.
c) Find the y-intercept and the gradient of your line of best fit.
d) Show that $y = ax^b$ may be written in the form

$$\log_{10} y = \log_{10} a + b \log_{10} x$$

e) Use your graph to estimate the values of a and b to 1 decimal place.

1.27 — Question 19. (*) The number of app downloads, N millions at time t months after launch may be modelled with the equation $N = at^b$. The data collected within the first 24 months is shown in the table below:

Time, t months	Downloads, N
3	12.1
5	24.7
8	50.3
10	65.9
18	170.2
24	249.1

a) Complete a table of values of $\log_{10} t$ and $\log_{10} N$, giving your answers to 3 decimal places.
b) Plot a graph of $\log_{10} N$ against $\log_{10} t$ and draw a line of best fit.
c) Use your graph to estimate the values of a and b to 1 decimal places.

Use your model to find

d) The amount of app downloads at 12 months.
e) The time it took to reach 200 million downloads.

1.27 — Question 20. (*) The mass of radioactive iodine, m grams after t days can be modelled with the exponential equation $m = ae^{bt}$ where a and b are constants. Data is collected every 10 days and is recorded in the table below:

Time, t days	Mass, m grams
0	50
10	20.9
20	8.9
30	3.6
40	1.7

a) Complete a table of values of $\ln m$ and t, giving your answers to 3 decimal places.
b) Plot $\ln m$ against t and draw a line of best fit on your graph.
c) Find the values of a and b
d) Use your model to predict the mass of iodine after 50 days since the start.

1.27 — Question 21. (*) Morgana has recorded how many minutes T she has spent exercising per week over the first 10 weeks of the year. She thinks the data may be modelled in the form $T = a \times t^b$ where t is the number of weeks since the beginning of the year. The data she collected is given in the table below:

Time, t weeks	Time spent exercising, T minutes
1	90
2	82
4	77
6	76
8	73
10	71

a) Plot a graph of $\log_{10} T$ against $\log_{10} t$ and draw a line of best fit on your graph.
b) Find the values of a and b and hence use your values of a and b to create a model in the form

$$T = a \times t^b$$

c) Interpret the meaning of the value a in the context of the model.
d) Use your model to find the amount of exercise she did 3 weeks after starting.
e) Comment on the validity of the model when $t = 0$
f) Comment on the long-term reliability of the model.

1.27 — Question 22. Matt is trying to find a model to represent the price of an item every month. He thinks it will be modelled in the form $y = ax^b$ where y is the price and x is the time passed in months that has passed.
He plotted $\log_{10} y$ against x, and found the y-intercept to be 2.4 and calculated the gradient as 0.6
He is unsure what to do next and suspects he has gone wrong.

a) State the mistake in his method.
b) Working with the graph he plotted, write an equation of y in terms of x with the information you have been given.

1.27 — Question 23. Dev has found a model for data that she has collected, but when she uses her model to make predictions the results are not what she expects.

x	y	$\ln x$
2	2300	7.74
4	2125	7.66
6	1975	7.59
8	1825	7.51
10	1675	7.24
12	1550	7.34

Gradient is -0.0395, intercept is 7.82

$$\ln y = -0.0395x + 7.82$$
$$y = 10^{-0.0395x+7.82}$$

a) Find the mistake in her working.
b) Find the equation of y in terms of x in the form

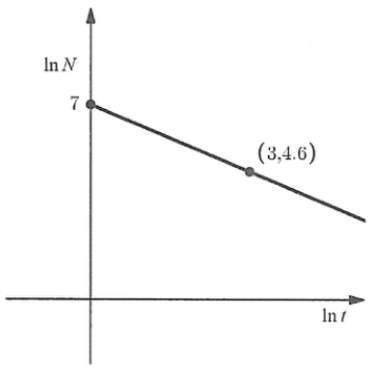
$$y = e^{px+q}$$

where p and q are constants to be found.

1.27 — Question 24. The number of cinema admissions N millions in the UK, may be modelled by the equation

$$N = a \times t^b$$

where t is the number of years since 1954. The plot below shows the graph of $\ln y$ against $\ln t$, with two of points on the line labelled.

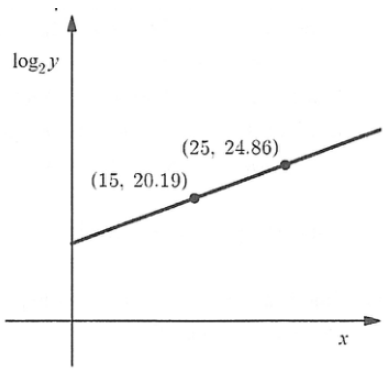


a) Find an equation for the line of best fit in terms of $\ln N$ and $\ln t$
b) Express N in terms of t
c) Use your model to predict the cinema admissions in 2020
d) Comment on the suitability of the model for predicting the number of cinema admissions in 2020

1.27 — Question 25. The semi-conductor industry uses Moore's Law to predict the number of transistors y in a microprocessor, modelled by

$$y = a \times 2^{bx}$$

where x is the number of years since 1975. The plot below shows the graph of $\log_2 y$ against x, with two points on the line labelled.



a) Find an equation for the line of best fit in terms of $\log_2 y$ and x
b) Express y in terms of x
c) With reference to the model, explain the meaning of the value of a in context
d) Use your model to predict the number of transistors in 2030

1.27 — Question 26. A content creator is monitoring the number of subscribers N to their channel, which may be modelled by

$$\log_{10} N = 0.238t + 1.975$$

where t is the number of subscribers t years after when the channel went live.

a) Show that $N = ab^t$ where a and b are constants to be found.
b) Interpret the meanings of the values a and b in the context of the model.
c) Use the model to calculate the total number of subscribers 10 years after the channel went live.

1.27 — Question 27. Zain leaves a cup of hot water to cool, he collects his data, and establishes a model for the temperature $T°\text{C}$ over time t minutes

$$\ln T = -0.0701t + 4.585$$

a) Determine the initial temperature of the hot water

b) State an equation for T in the form $T = ae^{bt}$ where a and b are constants.

1.27 — Question 28. It is found that the time it takes T days for a planet to orbit the sun may be given by

$$\ln T = 1.48 \ln d - 1.6$$

where d million kilometres is average distance of a planet from the Sun.

a) Find an equation for T in terms of d

Mars has an average distance of 228 million kilometres from the sun.

b) Using the model, find the number of days it will take Mars to orbit the sun, giving your answer to the nearest day.

It actually takes 687 days for Mars to orbit the sun.

c) Find the percentage error in the prediction made by the model.

1.27 — Question 29. The planet Saturn has many moons. The radius of each of the moons is given by r million km. The number of days it takes a moon to complete one orbit of Saturn is given by T days. They may be modelled by

$$r = a \times T^b$$

	Radius, r million km	Period, T days
Titan	1.22	15.9
Rhea	0.53	4.5

a) Show that the equation may be written in the form $\ln r = \ln a + b \ln T$
b) Given the data in the table, find the values of a and b
c) Sketch the graph of r against T, labelling points which represent Titan and Rhea.

Another moon of Saturn, Dione has a period of 2.7 days and a radius of 3.8×10^5 kilometres.

d) Use your model to predict the radius of the moon giving your answer to 3 significant figures.
e) Find the % error of the prediction by the model.

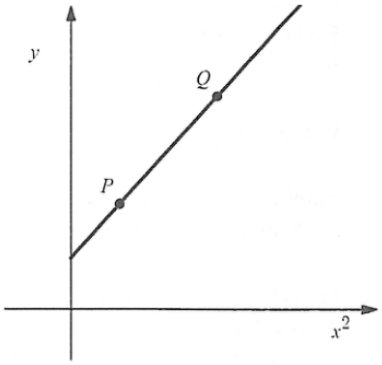
1.27 — Question 30. The life expectancy of a person aged y years, may be defined to be a function of the number of years they have left to live, x years. A model for this may be given by

$$\ln y = \ln b - ax$$

It is known that when $x = 5, y = 71$ and when $x = 55, y = 24$

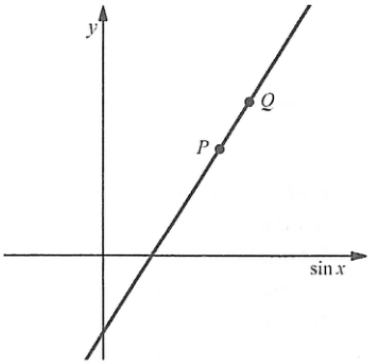
a) Find the value of a
b) Find the value of b
c) Use the model to find the life expectancy of someone aged 40
d) Find the age corresponding to a life expectancy of 25

1.27 — Question 31. The diagram below shows a straight-line graph of y against x^2 passing through $P(0.3, 5.41)$ and $Q(4.7, 8.49)$



Find an expression for y in the form $ax^b + k$ where a, b and k are constants to be found.

1.27 — Question 32. The diagram below shows a straight-line graph of y against $\sin x$ passing through $P(\frac{\pi}{2}, 5.3)$ and $Q(\frac{3\pi}{5}, 7.8)$



Write y in the form $a\sin(x) + b$ where a and b are constants to be found.